

function FORWARDNN($\mathbf{x}$, *network*) **returns** output sequence $\mathbf{y}$

$\mathbf{h}_0 \leftarrow 0$

for $i \leftarrow 1$ **to** LENGTH($\mathbf{x}$) **do**

$\mathbf{h}_i \leftarrow g(\mathbf{U}\mathbf{h}_{i-1} + \mathbf{W}\mathbf{x}_i)$

$\mathbf{y}_i \leftarrow f(\mathbf{V}\mathbf{h}_i)$

return $\mathbf{y}$